

Is DSP-free Pluggable Optical Module More Suitable for Optical Circuit Switching?

Chenxiao Zhang^{(1)*}, Han Wang^{(1)*}, Cen Wang⁽¹⁾, Daiki Soma⁽¹⁾, Yuta Wakayama⁽¹⁾,
Ryo Inohara⁽¹⁾, Takehiro Tsuritani⁽¹⁾

⁽¹⁾ KDDI Research, Inc., Fujimino, Saitama 356-8502, Japan, ch-zhang@kddi.com (*Equal Contribution)

Abstract *This paper presents the first comprehensive analysis of an LPO + OCS GPU network, from the optical to the application layer, including the impact of OCS reconfiguration. Stable 72-hour LLM pre-training was successfully demonstrated on this architecture. ©2026 The Author(s)*

Introduction

Inter-XPU communication has emerged as the dominant contributor to traffic in data center-scale networks [1]. As link rates scale toward 800G and 1.6T, both energy efficiency and switching capacity are increasingly becoming key constraints in the evolution of XPU interconnect architectures [2, 3]. To alleviate these bottlenecks, solutions such as Linear Pluggable Optics (LPO) have gained traction, in which DSP functionality is offloaded to the switching ASIC, thereby reducing per-port power consumption [4]. In parallel, Optical Circuit Switching (OCS) has been explored as a means to partially replace high-radix Electrical Packet Switches (EPSes) [5, 6]. OCS can increase switching capacity regardless of port rate while further reducing power consumption by more than $10\times$.

LPO and OCS are complementary rather than competing. Their hybridization provides a promising approach toward more energy-efficient and low-latency XPU networking. However, several unresolved issues and potential challenges remain to be thoroughly understood. Specifically, a) in the physical layer, when LPO transceivers are employed, the DSP is located remotely, placing more stringent demands on the integrity of analog signal transmission. As LPO optical signals traverse OCS devices, they may experience varying degrees of optical impairment. Meanwhile, b) in the network layer, irrespective of the network scheduling strategy, connections within the OCS must be periodically reconfigured, leading to link switchover events. Such operations impact the link up process differently for DSP-attached and DSP-free modules. Additionally, c) in the application layer, we are particularly concerned with how the stability and latency advantage of LPO + OCS impacts training job performance.

To this end, this paper presents an LPO + OCS platform tailored for XPU interconnects. Within this platform, to investigate the aforementioned questions, we conduct the following studies: (a) we evaluate the optical transmission characteristics through an 800G tester; (b) we compare the

link-up time of LPO and DSP-attached 800G modules under OCS reconfiguration; and (c) we perform uninterrupted pre-training over 72 hours and additionally assessed the impact of OCS switching on Job Completion Time (JCT). The results show that MEMS-based OCS introduces no observable impairment to the LPO receiving and the remote DSP. During OCS reconfiguration, an LPO exhibits a faster link-up time compared to a DSP-attached module. Furthermore, the LPO + OCS architecture reduces the LLM training JCT, while appropriately timed OCS reconfiguration has only a negligible impact on overall job performance.

LPO + OCS Platform

As illustrated in Fig. 1, for a) the physical layer, an 800G LPO-adaptive board is connected to the tester through copper cables. An LPO has two MPO-12 interfaces, which can be configured into different connection modes using breakout cables. Specifically, the 800G LPO is configured in a $2\times 400\text{GE}$ operating mode to enable both OCS-traversing and OCS-bypass connections. The tester is capable of measuring per-lane pre-FEC performance and symbol histogram, as well as the end-to-end latency.

For b) the network and c) the application layer evaluations, two 800G EPSes are used to emulate two cluster domains. The ASIC can support both DSP-attached modules and LPOs. Each cluster comprises two H100 GPUs. Inter-cluster traffic for each GPU is transmitted and received through a specific NIC. Each NIC provides a link capacity of 400 Gbps based on DR4. The EPSes are interconnected through 800G $2\times\text{DR4}$ LPO links. Each LPO is also connected to a MEMS-based 96×96 OCS fabric through breakout cables. To evaluate OCS reconfiguration, an additional LPO module is deployed on one of the EPSes; within each reconfiguration cycle, only one LPO maintains an active cross-connection with the corresponding LPO in the remote cluster.

Moreover, we deploy a standard pre-training job of the Qwen2.5-1.5B [7] LLM on this platform. The

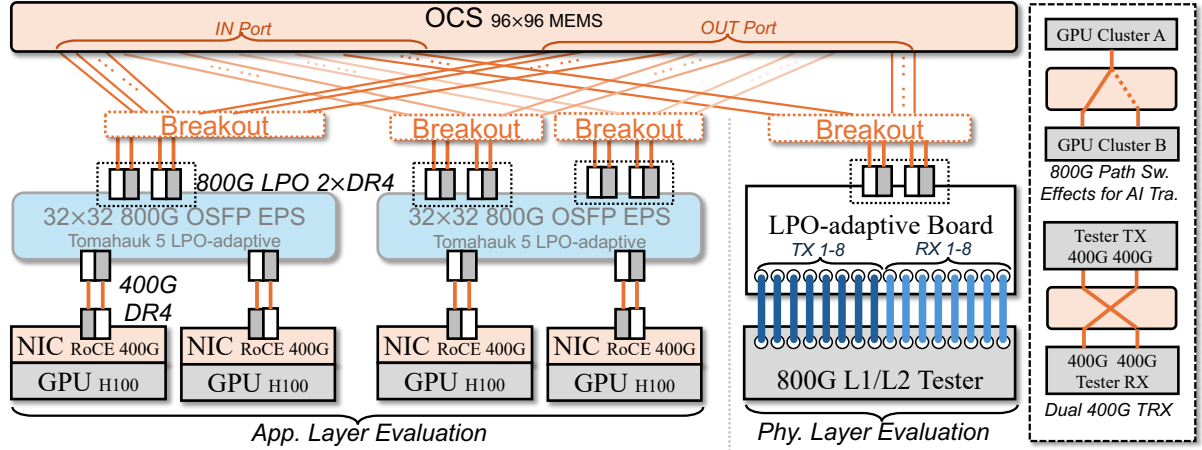


Fig. 1: LPO + OCS evaluation platform for both physical and application layer.

training dataset has a total size of 38 GB. This job fully utilizes the memory of four H100 GPUs. During training, a set of milestones (e.g., 100 GByte, and 400 GByte etc.) is defined, and the JCT is recorded whenever the cumulative traffic size reaches a milestone, enabling comparison across different experimental conditions. In addition, NIC transmit and receive traffic rate in bit-per-second is monitored at 100 ms intervals.

Optical Layer Characteristics

We first evaluate the symbol histograms of lanes through 1 to 8 for an 800G LPO operating in dual-400GE mode. The horizontal axis of the histogram represents the symbol amplitude, while the vertical axis indicates the number of symbol samples. A widening and flattening of the symbol peak distribution indicates the presence of noise, which introduces Inter-Symbol Interference (ISI) and correspondingly degrades the SNR. As shown in Fig. 2 (a) and (b), the impact of OCS depends on the intrinsic transmission quality of the LPOs. Specifically, for lanes 1 to 4, the lowest and highest symbol levels exhibit noticeable transmission impairments, which are further exacerbated by OCS. In contrast, for lanes 5 to 8, although some degradation is present, OCS appears to partially mitigate the original impairments of the LPO.

As shown in Fig. 2 (c), the impact of OCS on the

received pre-FEC BER of the LPO does not exhibit a consistent trend. OCS slightly increases the BER for only three out of the eight lanes, without any order-of-magnitude difference.

We further compare the end-to-end latency at the 400GE layer for LPO traversing and bypassing the OCS, as shown in Fig. 2 (d). The introduction of OCS results in an additional latency on the order of tens of nanoseconds. This increase is not consistent across different lanes, which may be attributed to the following factors: (i) the optical path delays within the OCS are not equal, and the overall latency increase is determined by the longest optical path; and (ii) the FEC convergence latency of the LPO is not stable, as reflected by an approximately 10 ns variation between RX1 and RX2. These observations suggest that latency management in LPO + OCS systems may require careful consideration of the internal structural consistency of the OCS. This characteristic may be negligible in non-uniform memory access XPU networks (e.g., scale-out and scale-across). However, in coherent memory XPU networks (e.g., scale-up), its impact on XPU idle time may require further investigation.

AI Job Performance

We conduct pre-training under three scenarios: LPO bypassing OCS, LPO traversing OCS, and

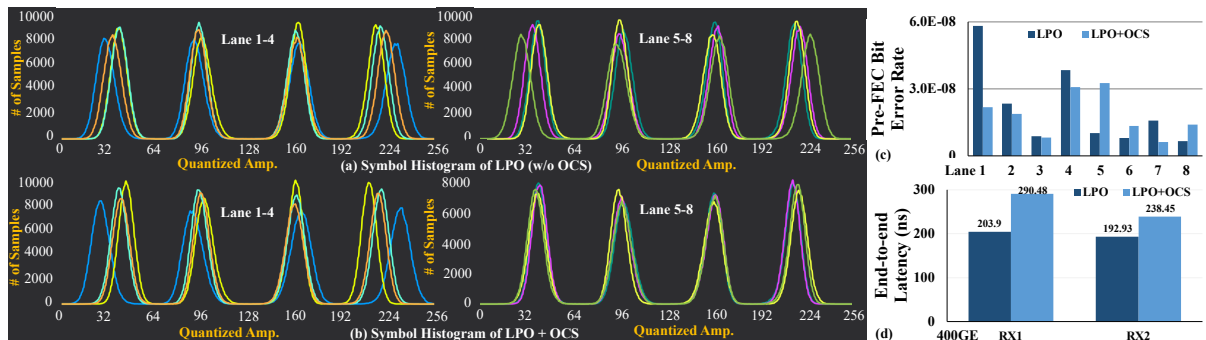


Fig. 2: Optical layer results: (a) and (b) symbol histogram, (c) pre-FEC BER, and (d) end-to-end latency measurement of LPO w/o OCS vs. LPO + OCS.

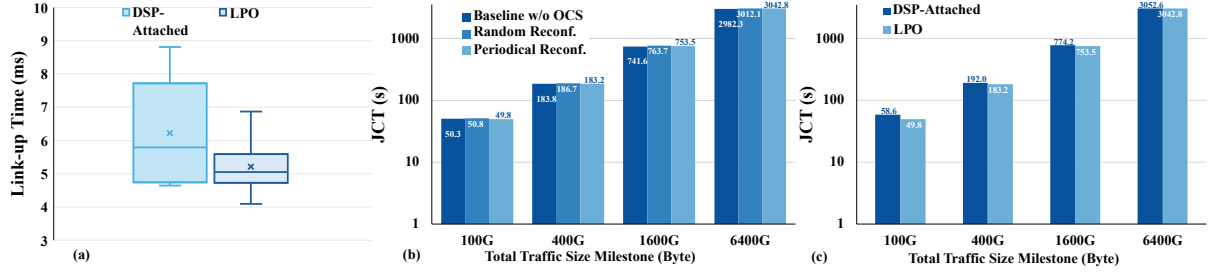


Fig. 3: Network and application layer evaluation results: (a) link up time; (b) impacts of OCS reconfigurations on JCT; (c) JCT of DSP-attached modules vs. LPOs + OCS.

DSP-attached modules traversing OCS. Each training run lasts up to 72 hours. To compare the JCTs across different experimental conditions, we record results at milestones ranging from 100 GB to 6400 GB of cumulative inter-GPU data transfer. The total data volume of the full pre-training process is extremely larger than 6400 GB, and as the JCT increases, the differences among experimental conditions become less statistically significant.

To reduce link-up latency, we disable the negotiation procedures and fix both the data rate and FEC mode. As shown in Fig. 3 (a), LPO exhibits significantly shorter and more stable link-up times compared to DSP-attached modules. Further reduction in link training time may be achieved if control over CTLE and TIA is enabled.

On the other hand, we observe the presence of zero-traffic windows during training, which typically correspond to GPU computation phases. In our experiments, these windows range from 100 ms to 400 ms. Even when accounting for the 25 ms cross-connect time of the OCS, optical path switching can be safely completed within these zero-traffic intervals. This indicates that, under certain conditions, LPO + OCS can feasibly support GPU direct-connect networking.

Due to the relatively stable periodicity of GPU computation phases, we design and compare the impact of random and periodic OCS reconfiguration strategies on JCT, as shown in Fig. 3(b). The baseline corresponds to the JCT measured when LPO bypasses the OCS. When statistically meaningful results are available, neither reconfiguration strategy shows a significant impact on JCT. However, random reconfiguration leads to frequent training interruptions, preventing the collection of JCT at certain milestones. In contrast, periodic reconfiguration results in fewer interruptions.

Further analysis indicates that these interruptions occur when OCS reconfiguration does not align with zero-traffic windows, but instead coincides with periods of high traffic transmission. When a link interruption occurs during high-rate NIC transmission, RoCE is unable to perform effective flow control and packet loss is not permissible. As a result, NIC buffers may overflow, ultimately leading to network failures on the NIC side.

It should be noted that the job employs a ring all-reduce parallelism, which does not require actual topology changes during training. However, the aforementioned results indicate that even for parallelism with dynamic topology requirements (e.g., recursive doubling or more complex combinations of data and model parallelism), it is feasible to design a traffic-adaptive OCS reconfiguration strategy (with awareness of zero-traffic windows) that matches to the topology demands of a specific parallelism. Such an approach can provide isolated high-capacity paths for inter-GPU communication, thereby avoiding traffic contention and congestion.

Finally, we compare the JCTs of LPOs and DSP-attached modules when traversing the OCS, without operating OCS reconfiguration. As shown in Fig. 3(c), the LPO + OCS achieves a reduction in JCT of approximately tens of seconds. This acceleration is likely due to the end-to-end latency advantage of LPOs, and it becomes less pronounced as the overall training duration increases.

Conclusion

This work experimentally validated the performance of an 800G 2×DR4 LPO + MEMS OCS architecture, using the 800G tester and H100 GPU clusters. Optical transmission measurements show that the introduction of OCS does not significantly degrade LPO RX performance. At the network link layer, OCS introduces variability in end-to-end latency; during OCS reconfiguration, LPO exhibits shorter link-up times compared to DSP-attached modules. At the application layer, under RoCE, the observed latency variability has no significant impact on JCT, while the reduced link-up time enables properly designed OCS reconfiguration strategies to operate without noticeably affecting JCT. Nevertheless, if OCS switching occurs during periods of high traffic, it can directly interrupt the training job. Furthermore, we successfully demonstrated stable 72-hour LLM pre-training across two GPU clusters on the proposed LPO + OCS architecture.

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